Lab 7: Fuzzing for Input Validation Bugs

Name: Akshatha P | SRN: PES2UG24CS048

I used Hypothesis to fuzz the three functions in scholarship_awards.py, saw them crash, fixed them, and re-ran the tests. Screenshots are below.

1. Running the original script

I ran the original script with normal inputs. It works fine when the input is clean, so the bugs only show up with odd inputs.

```
PS C:\Users\aksha\L6_L7_44_scholarship_awards\Lab 7> python scholarship_awards.py
==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:83
Category and Marks: ('Merit', 83.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 55,78,23
Average Marks: 52

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 4000|400
Payable Award: 4320.0
PS C:\Users\aksha\L6_L7_44_scholarship_awards\Lab 7>
```

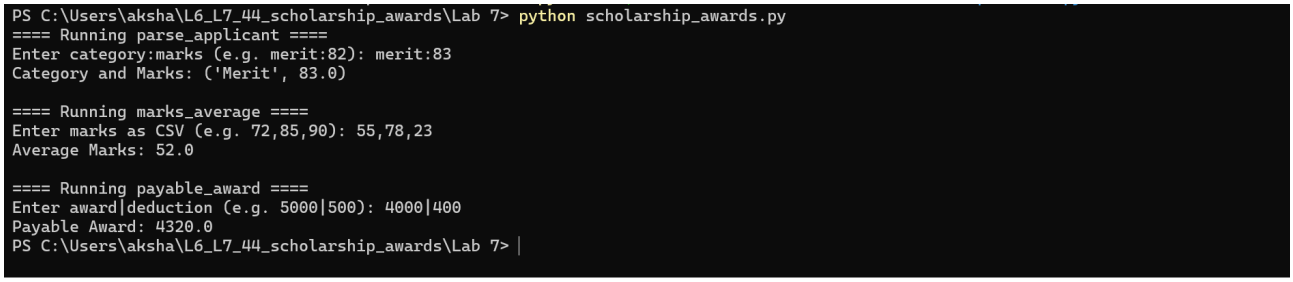
2. Fuzz tests on the original code

I wrote Hypothesis tests for all three functions. Hypothesis fed them things like None, empty strings and a lone '|', and all three functions crashed (3 failed, 3 passed).

```
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\aksha\L6_L7_44_scholarship_awards\Lab 7\test_scholarship_awards.py", line 29, in test_payable_award_no_crash
    |     payable_award(s)
    |   File "C:\Users\aksha\L6_L7_44_scholarship_awards\Lab 7\scholarship_awards.py", line 10, in payable_award
    |     award, deduction = text.split('|')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\aksha\L6_L7_44_scholarship_awards\Lab 7\test_scholarship_awards.py", line 31, in test_payable_award_no_crash
    |     assert False, f"payable_award crashed with: {e!r} for input {s!r}"
    | AssertionError: payable_award crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
    | assert False
    | Failing test case: test_payable_award_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================================== short test summary info ============================================
FAILED test_scholarship_awards.py::test_parse_applicant_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_scholarship_awards.py::test_marks_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_scholarship_awards.py::test_payable_award_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================================ 3 failed, 3 passed in 6.80s ============================================
PS C:\Users\aksha\L6_L7_44_scholarship_awards\Lab 7>
```

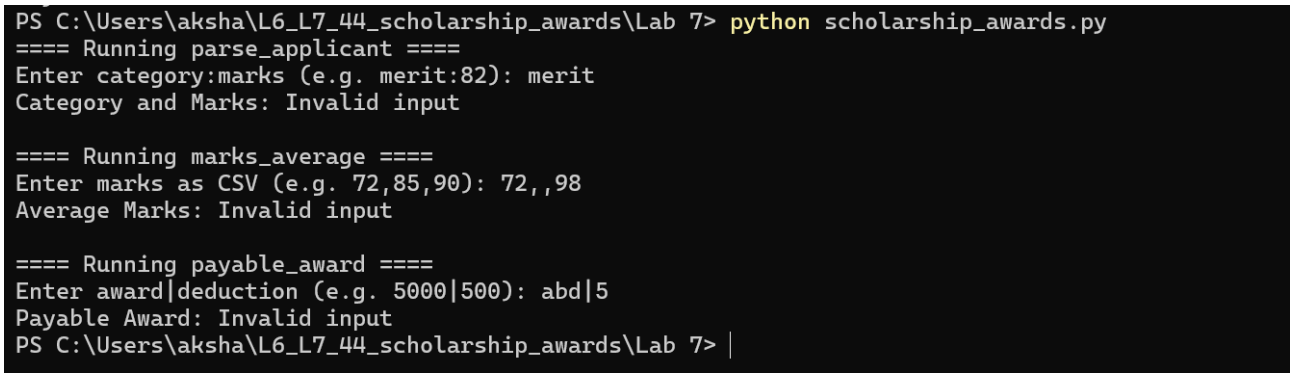
3. Fixed script, normal input

I added input checks to each function: type check, correct number of parts, valid numbers and ranges. Normal input still works. The average now shows 52.0 because I replaced floor division with proper division.



4. Fixed script, bad input

Bad input like 'merit', '72,,98' and 'abd|5' now prints 'Invalid input' instead of crashing.



5. Fuzz tests on the fixed code

Re-running the same tests on the fixed script, all 6 pass.

